

Consider the following endpoints created with a working set up of flask-security and flask-sqlalchemy.

```
@app.route('/api/public')
def public_endpoint():
    return jsonify({'message': 'Public access', 'status': 'success'})

@app.route('/api/protected')
@auth_required('token')
def protected_endpoint():
    return jsonify({'message': 'Protected access', 'status':
'authenticated'})

@app.route('/api/admin')
@auth_required('token')
@roles_required('admin')
def admin_endpoint():
    return jsonify({'message': 'Admin access', 'status': 'authorized'})
```

Three requests are made to this application:

1. GET /api/public (no authentication headers)
2. GET /api/protected (no authentication headers)
3. GET /api/admin with valid authentication token but user has role (not 'admin')

What will be the response status codes for these requests, respectively?

Options :

6406534707719. ✖ 200, 200, 200

6406534707720. ✔ 200, 401, 403

6406534707721. ✖ 200, 401, 401

6406534707722. ✖ 401, 401, 403

MLT

Section Id :	64065399163
Section Number :	7
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	15
Number of Questions to be attempted :	15
Section Marks :	50
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No

Enable Mark as Answered Mark for Review and Clear Response : No
Section Maximum Duration : 0
Section Minimum Duration : 0
Section Time In : Minutes
Maximum Instruction Time : 0
Sub-Section Number : 1
Sub-Section Id : 640653226007
Question Shuffling Allowed : No

Question Number : 152 Question Id : 6406531405167 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING TECHNIQUES (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406534707723. ✓ YES

6406534707724. ✗ NO

Sub-Section Number : 2
Sub-Section Id : 640653226008
Question Shuffling Allowed : Yes

Question Number : 153 Question Id : 6406531405168 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider a dataset $\{x_1, x_2, \dots, x_n\}$, where each $x_i \in \mathbb{R}^5$. Principal Component Analysis (PCA) is applied to this dataset and all the five principal components are retained. Which of the following statements is/are correct?

Options :

6406534707725. ✗ PCA has reduced the dimensionality of the dataset because the combinations of features are de-correlated.

6406534707726. ✓ PCA has not reduced the dimensionality of the dataset because all principal components are retained.

6406534707727. ✗ PCA always reduces the dimensionality of the dataset regardless of the number of components retained.

6406534707728. ✗ PCA can never be used for dimensionality reduction.

Question Number : 154 Question Id : 6406531405170 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider the following kernel function:

$$k : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$k([x_1, x_2]^T, [y_1, y_2]^T) = 1 + x_1 y_1 + x_2 y_2 + x_1^2 y_1^2 x_2 y_2 + x_1 y_1 x_2^2 y_2^2$$

Find the appropriate transformation mapping ϕ for the given kernel.

Options :

6406534707733. ✖ $\phi([x_1, x_2]^T) = [1, x_1 x_2, x_1^2 x_2^2]$

6406534707734. ✖ $\phi([x_1, x_2]^T) = [1, x_1, x_2, x_1^2 x_2^2]$

6406534707735. ✔ $\phi([x_1, x_2]^T) = [1, x_1, x_2, x_1^2 x_2, x_1 x_2^2]$

6406534707736. ✖ $\phi([x_1, x_2]^T) = [1, x_1, x_2, x_1 x_2, x_1^2 x_2^2]$

Sub-Section Number :

3

Sub-Section Id :

640653226009

Question Shuffling Allowed :

Yes

Question Number : 155 Question Id : 6406531405169 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider clustering 1D data with a mixture of 2 Gaussians using the EM algorithm. You are given the data points $x_1 = 1, x_2 = 10, x_3 = 20$.

Suppose the output of the E-step is as follows:

$$\lambda_1^1 = 1, \lambda_2^1 = 0$$

$$\lambda_1^2 = 0.4, \lambda_2^2 = 0.6$$

$$\lambda_1^3 = 0.1, \lambda_2^3 = 0.9,$$

where λ_c^i denotes the probability of the i -th data point to be in cluster C . After performing the M-step for the means μ_1 and μ_2 , what are the updated means?

Options :

6406534707729. ✖ $\mu_1 = 0.5, \mu_2 = 0.5$

6406534707730. ✖ $\mu_1 = 0.5, \mu_2 = 1.6$

6406534707731. ✖ $\mu_1 = 4.66, \mu_2 = 4.66$

6406534707732. ✔ $\mu_1 = 4.66, \mu_2 = 16$

Sub-Section Number :

4

Sub-Section Id :

640653226010

Question Shuffling Allowed :

Yes

Question Number : 156 Question Id : 6406531405171 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Consider a linearly separable binary classification problem in which the data-points are in $\mathbb{R}^{100}$. The training dataset has 100 data-points, 40 from the positive class and 60 from the negative class. Now, consider a hard-margin linear SVM trained on this dataset. How many constraints does the primal problem have? Select the most appropriate answer.

Options :

6406534707737. ✖ 40

6406534707738. ✖ 60

6406534707739. ✔ 100

6406534707740. ✖ 1

Sub-Section Number :

5

Sub-Section Id :

640653226011

Question Shuffling Allowed :

Yes

Question Number : 157 Question Id : 6406531405172 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which among the following is TRUE for k -NN algorithm? Select all that apply.

Options :

6406534707741. ✔ 1-nearest neighbour is sensitive to outliers.

6406534707742. ✖ k -NN can only be applied to a classification problem.

6406534707743. ✖ As k increases, the model is more likely to overfit.

6406534707744. ✔ The bias increases with the increase in the value of k .

6406534707745. ✖ The variance increases with the increase in the value of k .

Question Number : 158 Question Id : 6406531405174 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Assume that perceptron algorithm is applied to a data set in which the maximum lengths of the data points is 5 and the value of margin (γ) of the optimal separator is 1. If the algorithm has made 5 mistakes at some point of the execution of the algorithm, which of the following can be a valid squared length of the weight vector obtained in the 6th iteration?

Options :

6406534707750. ✓ 30

6406534707751. ✓ 120

6406534707752. ✓ 50

6406534707753. ✗ 160

Question Number : 159 Question Id : 6406531405175 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Let w^*, ξ^* be the optimal primal solutions, and α^*, β^* be the optimal dual solutions of the soft-margin SVM problem. If $\alpha_i^* = 0$, select the correct options.

Options :

6406534707754. ✗ The bribe paid by the i^{th} data point is equal to C .

6406534707755. ✓ The bribe paid by the i^{th} data point is equal to 0.

6406534707756. ✗ i^{th} data point lies on the supporting hyper-planes.

6406534707757. ✓ w^* classifies the i^{th} data point correctly.

Sub-Section Number :

6

Sub-Section Id :

640653226012

Question Shuffling Allowed :

Yes

Question Number : 160 Question Id : 6406531405173 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

With respect to the Lloyd's algorithm, choose the correct statements:

Options :

6406534707746. ✓ The partition configurations cannot repeat themselves.

6406534707747. ✖ After doing the reassignments (consider at least one point reassigned to the new cluster), we might get the same means for all clusters.

6406534707748. ✔ Objective function after making the re-assignments strictly reduces.

6406534707749. ✖ Objective function after making the re-assignments strictly increases.

Sub-Section Number : 7
Sub-Section Id : 640653226013
Question Shuffling Allowed : Yes

Question Number : 161 Question Id : 6406531405176 Question Type : SA
Correct Marks : 3

Question Label : Short Answer Question

Consider a naive Bayes classification problem with three binary features f_1, f_2, f_3 for a binary classification problem. We have a total of 8 training samples given in the table below:

Sample	f_1	f_2	f_3	y
x_1	1	1	0	0
x_2	1	0	0	0
x_3	1	1	0	0
x_4	1	0	0	0
x_5	1	0	0	1
x_6	1	1	0	1
x_7	1	0	0	1
x_8	1	1	0	1

If a test point has label 0, what is the probability of $x_{test} = [1 \ 1 \ 1]^T$? Assume that the Laplacian smoothing is done, (i.e., add two pseudo training examples $x_{pseudo}^0 = [1 \ 1 \ 1]^T$ for class 0 and $x_{pseudo}^1 = [1 \ 1 \ 1]^T$ for class 1). Enter the answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.10 to 0.14

Sub-Section Number : 8
Sub-Section Id : 640653226014
Question Shuffling Allowed : Yes

Question Number : 162 Question Id : 6406531405177 Question Type : SA
Correct Marks : 4

Question Label : Short Answer Question

Consider a binary classification problem with three data points, denoted by $\{(x_i, y_i)\}$, where x_i represents a one-dimensional feature and y_i is the corresponding class label, as given below:

$$\{(1, +1), (2, -1), (3, +1)\}$$

Each data point gets uniform initial weights $w_1^{(0)} = w_2^{(0)} = w_3^{(0)} = \frac{1}{3}$, where $w_i^{(t)}$ denotes the weight of i -th data point for t^{th} iteration. Following are the two decision stumps given to us:

$$h_1(x) = \begin{cases} +1, & \text{if } x \leq 2 \\ -1, & \text{if } x > 2 \end{cases}, \quad h_2(x) = \begin{cases} +1, & \text{if } x \leq 1 \\ -1, & \text{if } x > 1 \end{cases}.$$

We run the AdaBoost algorithm for one iteration by selecting the decision stump with the lowest training error. Then, find the value of $w_1^{(1)} + w_2^{(1)}$. Enter the answer correct to one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.5

Question Number : 163 Question Id : 6406531405178 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

Suppose we are solving a binary classification task using a neural network with the following architecture:

- Input: $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- Hidden layer: 2 neurons (ReLU activation)
- Output layer: 1 neuron (sigmoid activation)

The weight parameters of the network are given as:

$$W^{(1)} = \begin{bmatrix} w_1^{(1)} & w_2^{(1)} \\ w_1^{(2)} & w_2^{(2)} \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}, \quad W^{(out)} = \begin{bmatrix} w_1^{(out)} & w_2^{(out)} \end{bmatrix}^T = \begin{bmatrix} 1 & 0 \end{bmatrix}^T,$$

where $w_i^{(j)}$ represents the weight associated with the j -th neuron for the i -th input feature. Compute the cross-entropy loss L . Write the answer to correct to three decimal places.

Loss function: $L = -[y \log_e(\hat{y}) + (1 - y) \log_e(1 - \hat{y})]$, where $\hat{y}$ is the network output and $y = 1$ is the true label.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.69 to 0.696

Sub-Section Number :

9

Sub-Section Id :

640653226015

Question Shuffling Allowed :

No

Question Id : 6406531405179 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (164 to 165)

Question Label : Comprehension

Consider a linear regression problem with design matrix $X \in \mathbb{R}^{d \times n}$ and target vector $Y \in \mathbb{R}^{n \times 1}$, where n is the number of data points and d is the number of features. Let $w \in \mathbb{R}^{d \times 1}$ denote the vector of regression coefficients.

$$X = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad Y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 164 Question Id : 6406531405180 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Calculate $\hat{w}_{ML}$ and $\hat{w}_{Ridge}$ for the regularization parameter $\lambda = 1$.

Options :

6406534707761. ✖ $\hat{w}_{ML} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad \hat{w}_{Ridge} = \begin{bmatrix} 0.20 \\ 1.49 \end{bmatrix}$

6406534707762. ✖ $\hat{w}_{ML} = \begin{bmatrix} 0.20 \\ 1.49 \end{bmatrix}, \quad \hat{w}_{Ridge} = \begin{bmatrix} -1.5 \\ 3 \end{bmatrix}$

6406534707763. ✖ $\hat{w}_{ML} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad \hat{w}_{Ridge} = \begin{bmatrix} -1.5 \\ 3 \end{bmatrix}$

6406534707764. ✓ $\hat{w}_{ML} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad \hat{w}_{Ridge} = \begin{bmatrix} 0.37 \\ 1.34 \end{bmatrix}$

Question Number : 165 Question Id : 6406531405181 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Repeat the ridge regression coefficients for $\lambda = 10$. How do the estimate for w change as λ increases?

Options :

6406534707765. ✓ The coefficients shrinks towards zero.

6406534707766. ✗ The coefficients increases in magnitude.

6406534707767. ✗ The coefficients remain unchanged.

Question Id : 6406531405182 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (166 to 167)

Question Label : Comprehension

A medical researcher collects data to predict whether a patient has a certain disease based on the variables $X_1 = \text{age (in years)}$, and $X_2 = \text{cholesterol level (in mg/dL)}$. Let $Y = 1$ denotes the patients with the disease. A logistic regression model is fit which produces the estimates of $w = \begin{bmatrix} w_1 & w_2 \end{bmatrix}^T$ as:

$$\hat{w}_1 = -0.05, \quad \hat{w}_2 = 0.02.$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 166 Question Id : 6406531405183 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

What is the probability that a 60-year-old patient with a cholesterol level of 200 mg/dL has the disease? Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.71 to 0.75

Question Number : 167 Question Id : 6406531405184 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

How high should the cholesterol level be for the patient mentioned in the previous question to have a 50% chance of being diagnosed with the disease?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

150

Sub-Section Number : 10

Sub-Section Id : 640653226016

Question Shuffling Allowed : No

Question Id : 6406531405185 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (168 to 169)

Question Label : Comprehension

Consider the datapoints $(-2, -2)$ with label $+1$ and $(2, 2)$ with label -1 . A hard margin SVM is applied on this dataset. Answer the given subquestions:

Sub questions

Question Number : 168 Question Id : 6406531405186 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following is the decision boundary for the given dataset?

Options :

6406534707770. ✖ $x + y \geq 0$

6406534707771. ✔ $-x - y \geq 0$

6406534707772. ✖ $x - y \geq 0$

6406534707773. ✖ $y - x \geq 0$

Question Number : 169 Question Id : 6406531405187 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

The solution to the dual problem is given by $\alpha^* = \begin{bmatrix} \frac{1}{16} & \frac{1}{16} \end{bmatrix}^T$. Find the width of the widest margin for this dataset. Enter the answer correct to two decimal places.

Note: Width of the widest margin is given by $\frac{2}{||w||}$.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
5.64 to 5.68

MLP

Section Id :	64065399164
Section Number :	8
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	21
Number of Questions to be attempted :	21
Section Marks :	50
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653226017
Question Shuffling Allowed :	No